

Model Analysis Report

Generated: February 26, 2026

Dataset Information

Number of timesteps: 136214

Spatial grid: 50 latitudes × 24 MLTs

Time range: 2019-11-01 00:00:00 to 2024-01-31 23:56:00

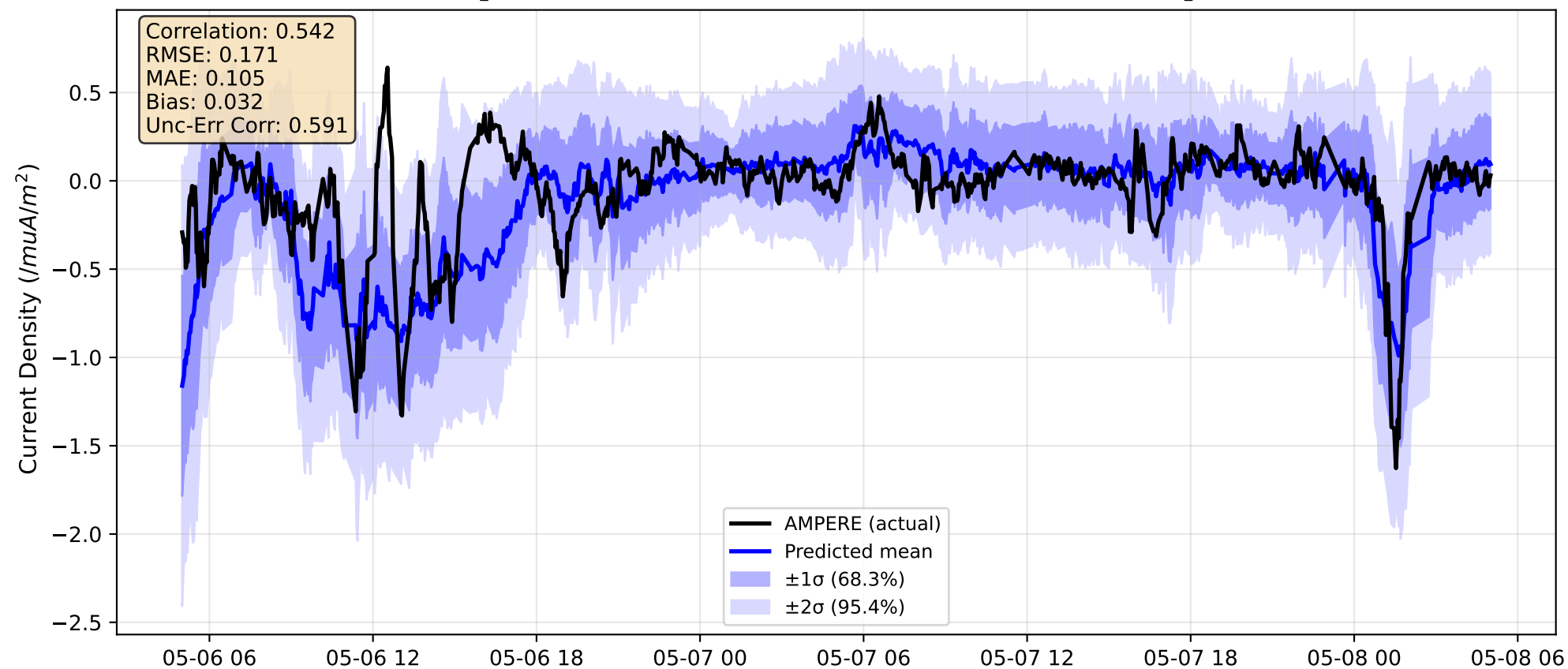
Uncertainty available: Yes

Analysis location: Lat=70°, MLT 4-8

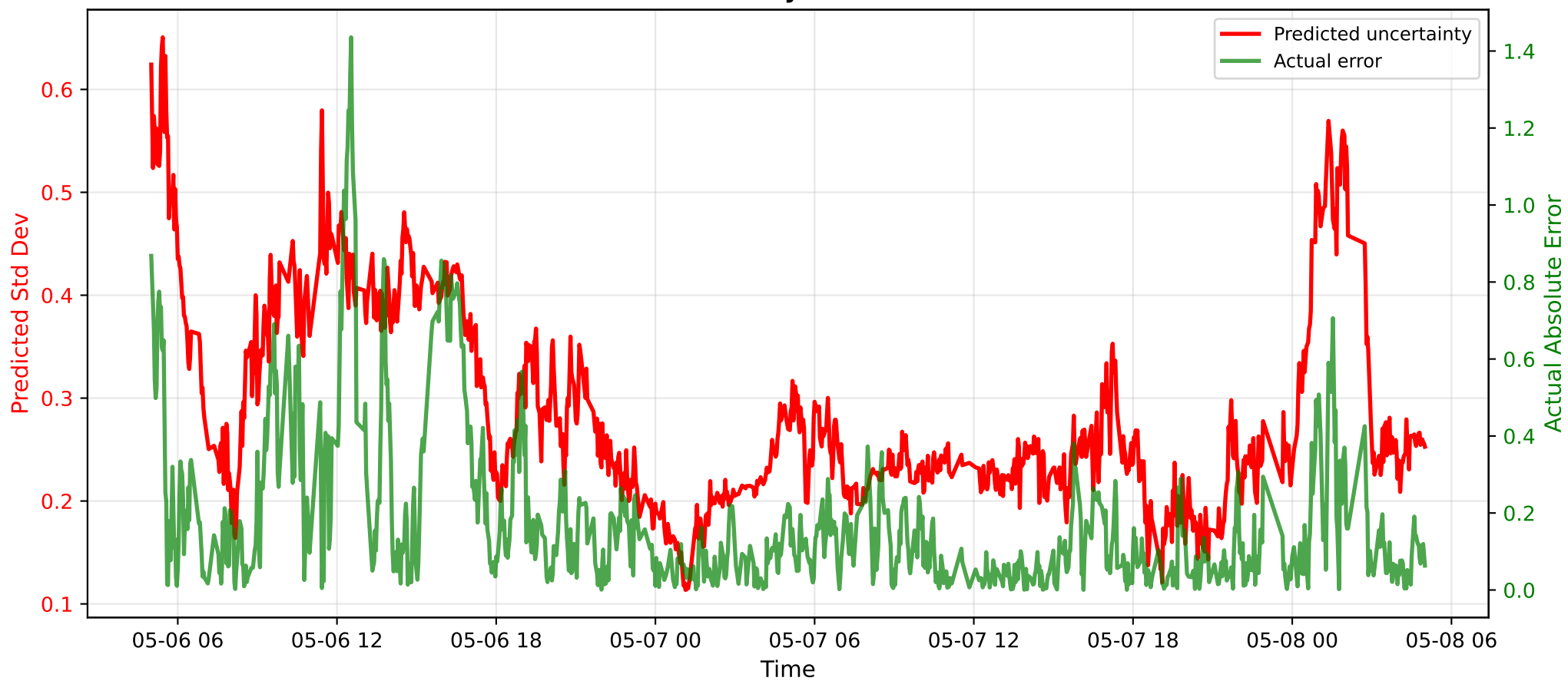
Plot time range: 2023-05-04 05:00:00 to 2023-05-08 05:00:00

*This report contains comprehensive validation metrics
for geomagnetic model predictions*

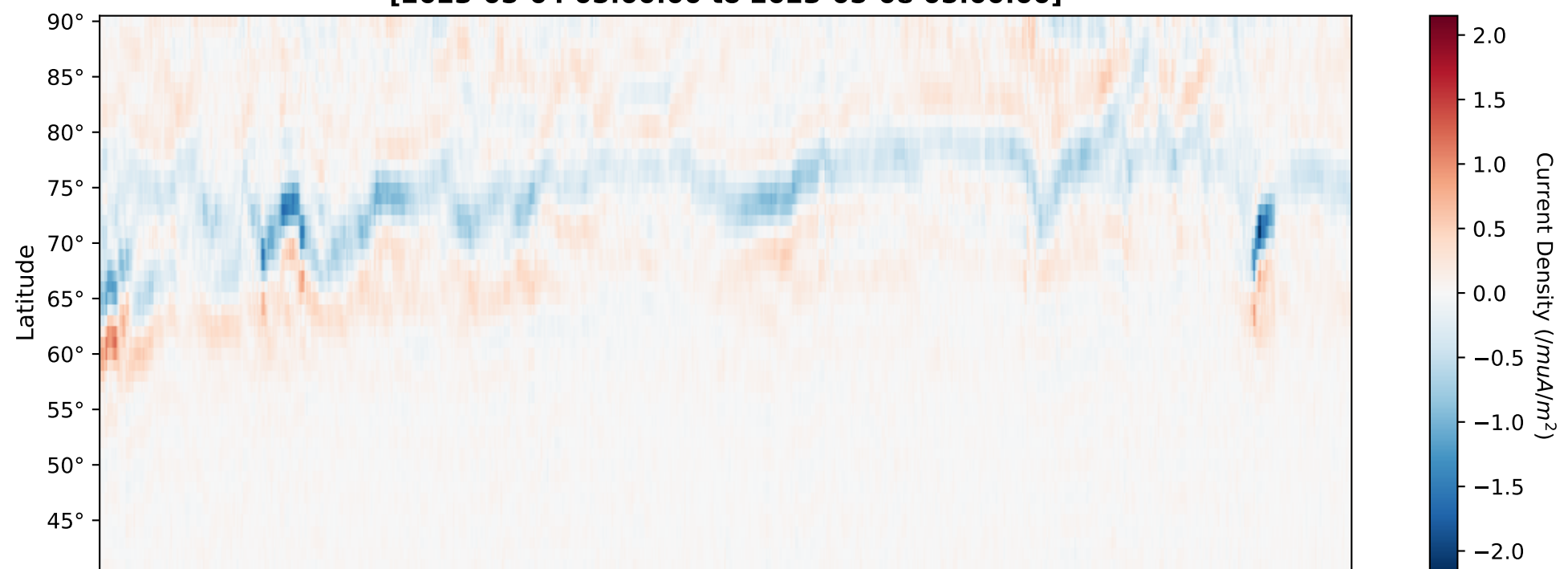
Time Series with Uncertainty (Lat=70°, MLT 4-8)
[2023-05-04 05:00:00 to 2023-05-08 05:00:00]



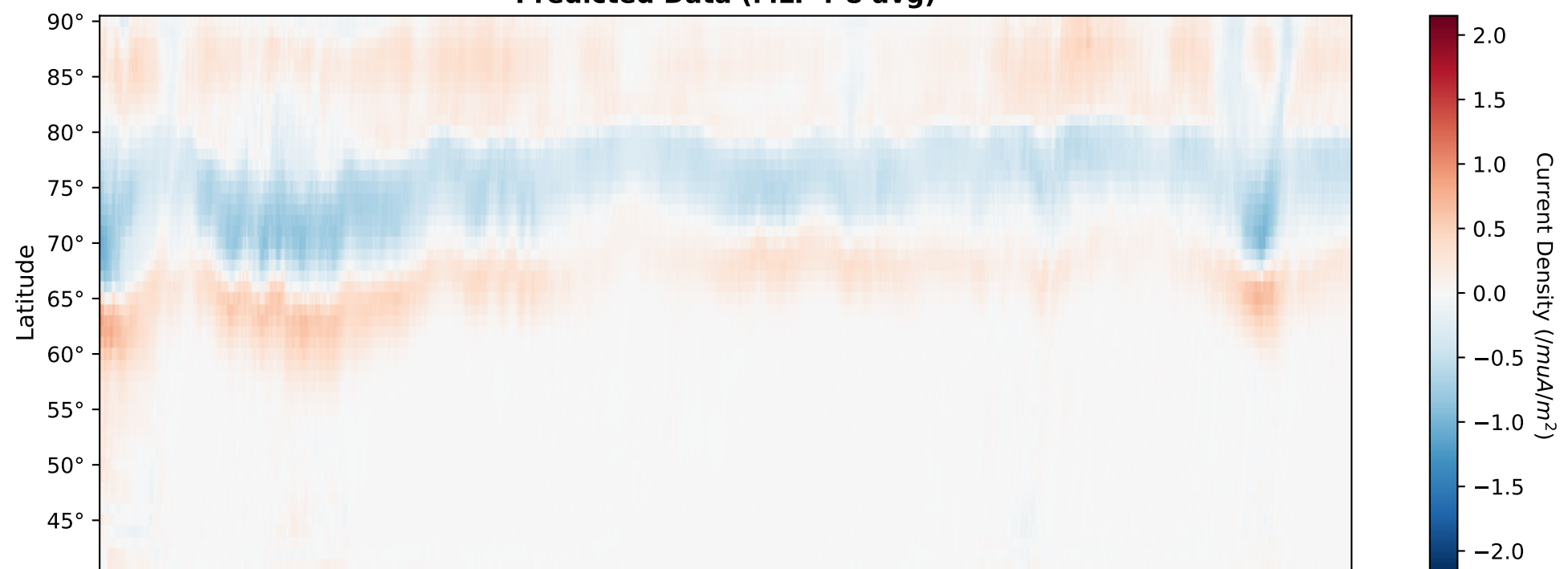
Uncertainty Evolution



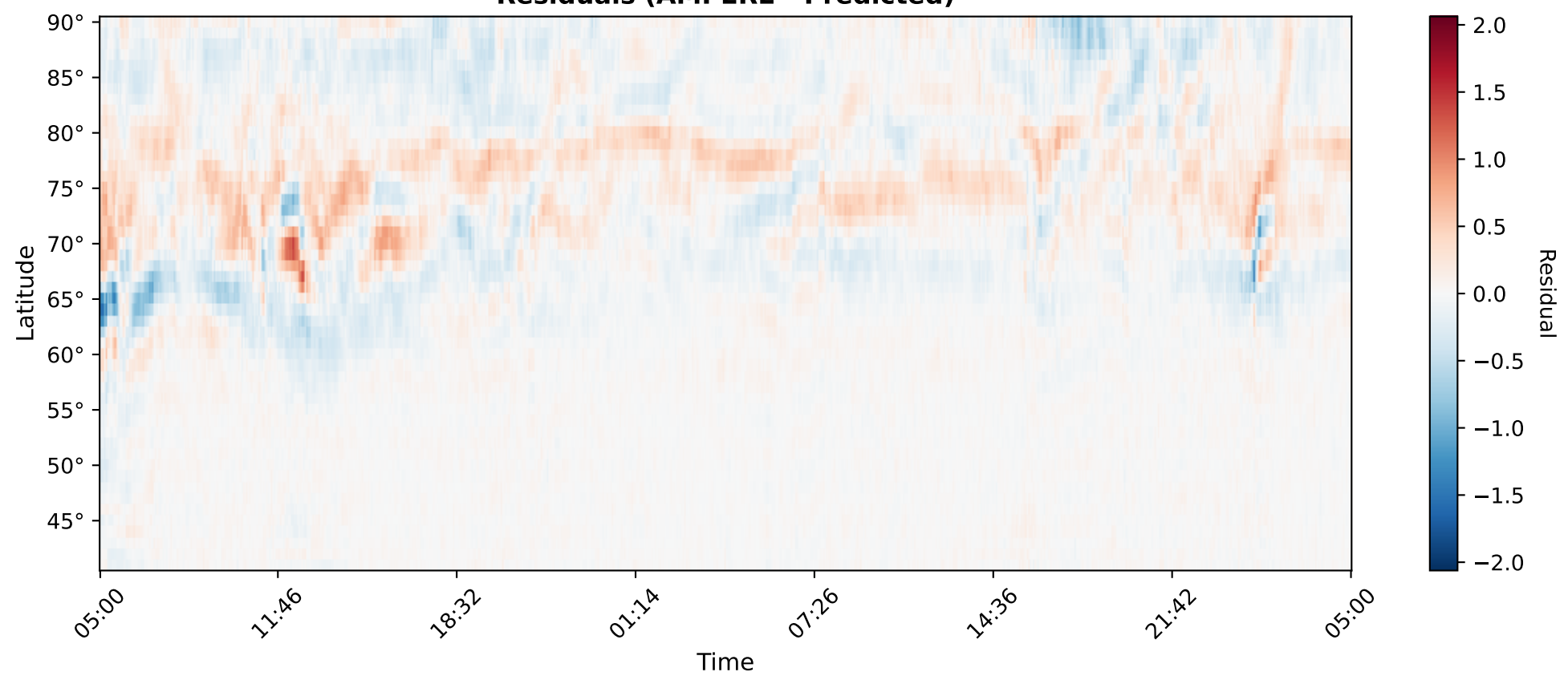
AMPERE Data (MLT 4-8 avg)
[2023-05-04 05:00:00 to 2023-05-08 05:00:00]



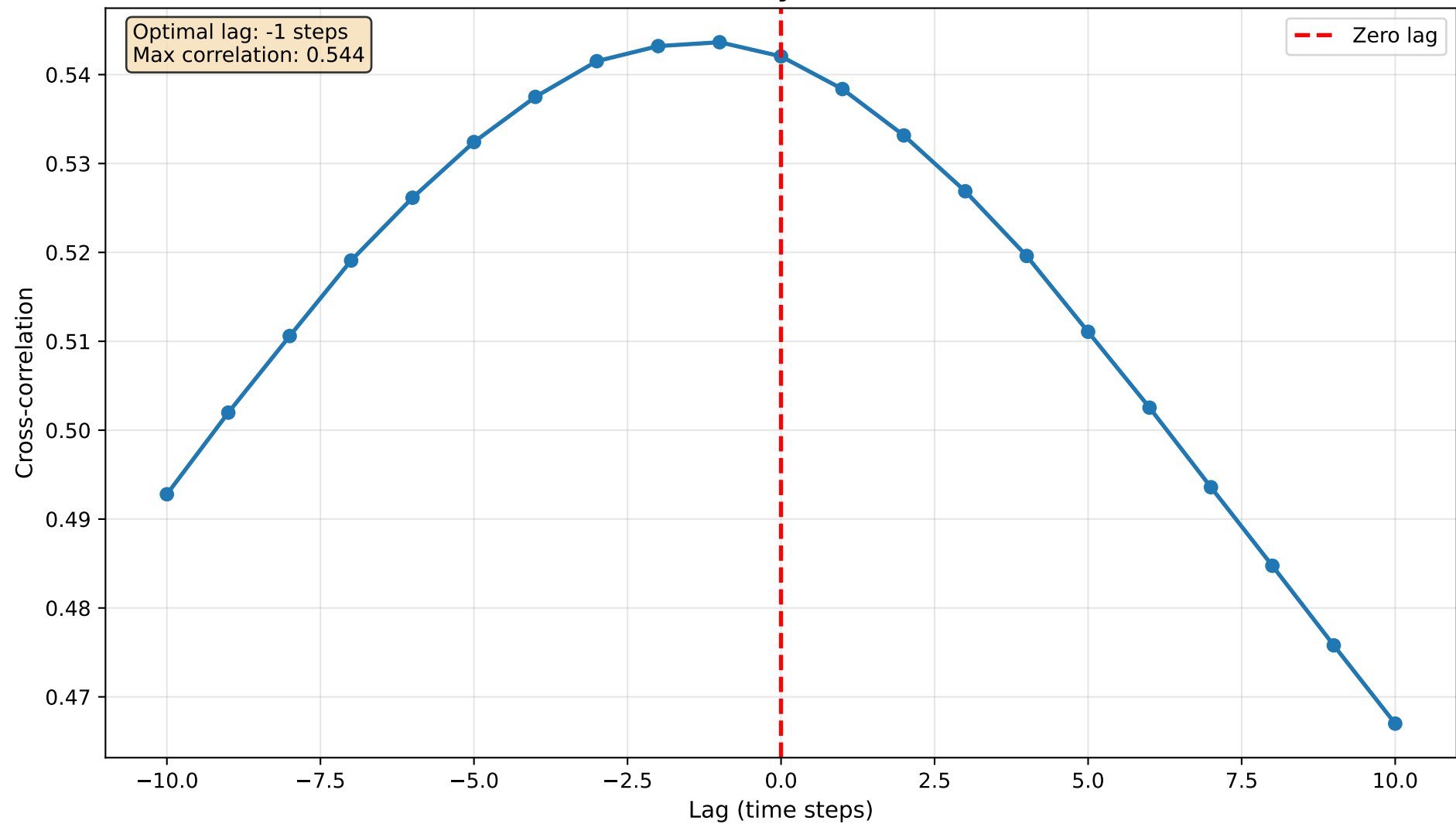
Predicted Data (MLT 4-8 avg)



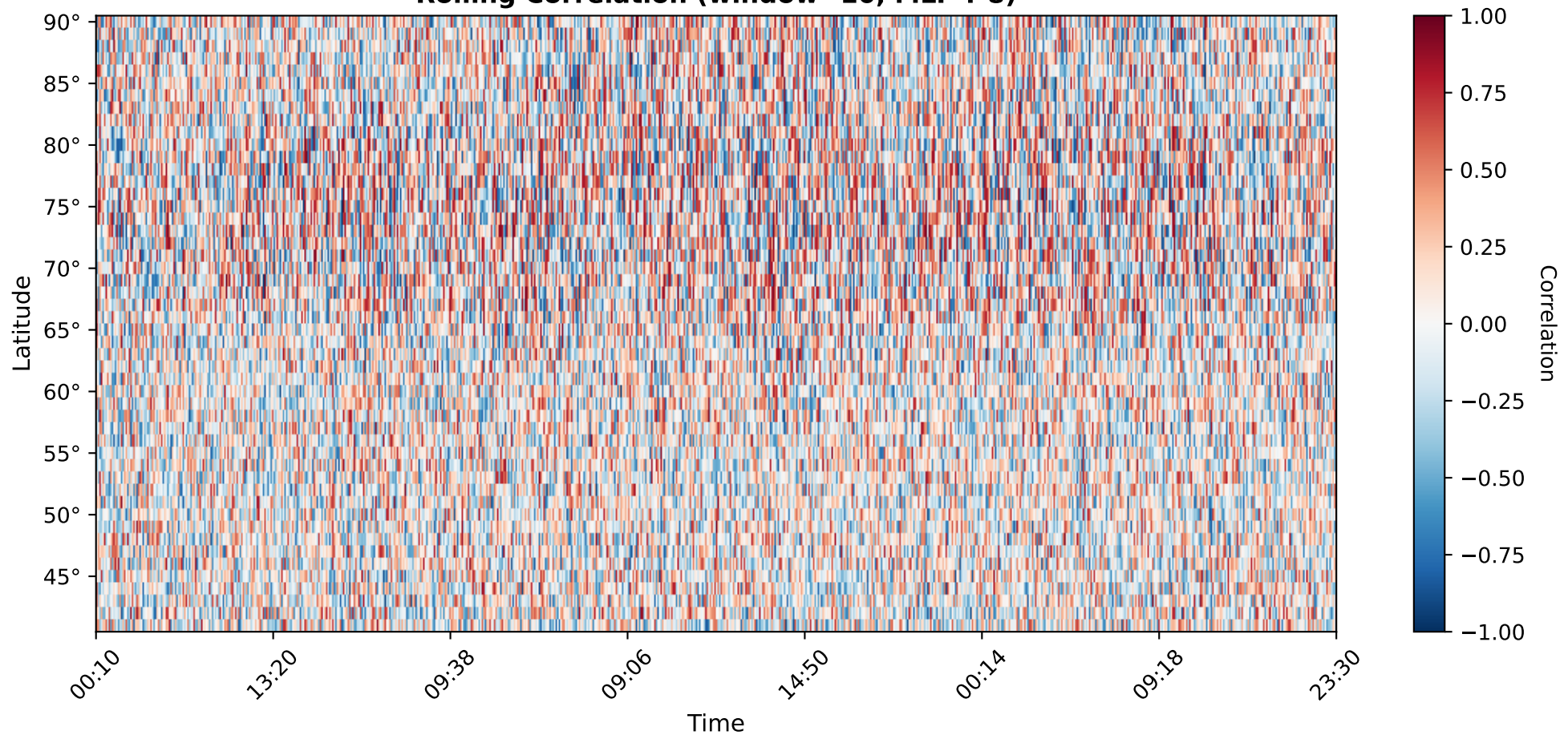
Residuals (AMPERE - Predicted)



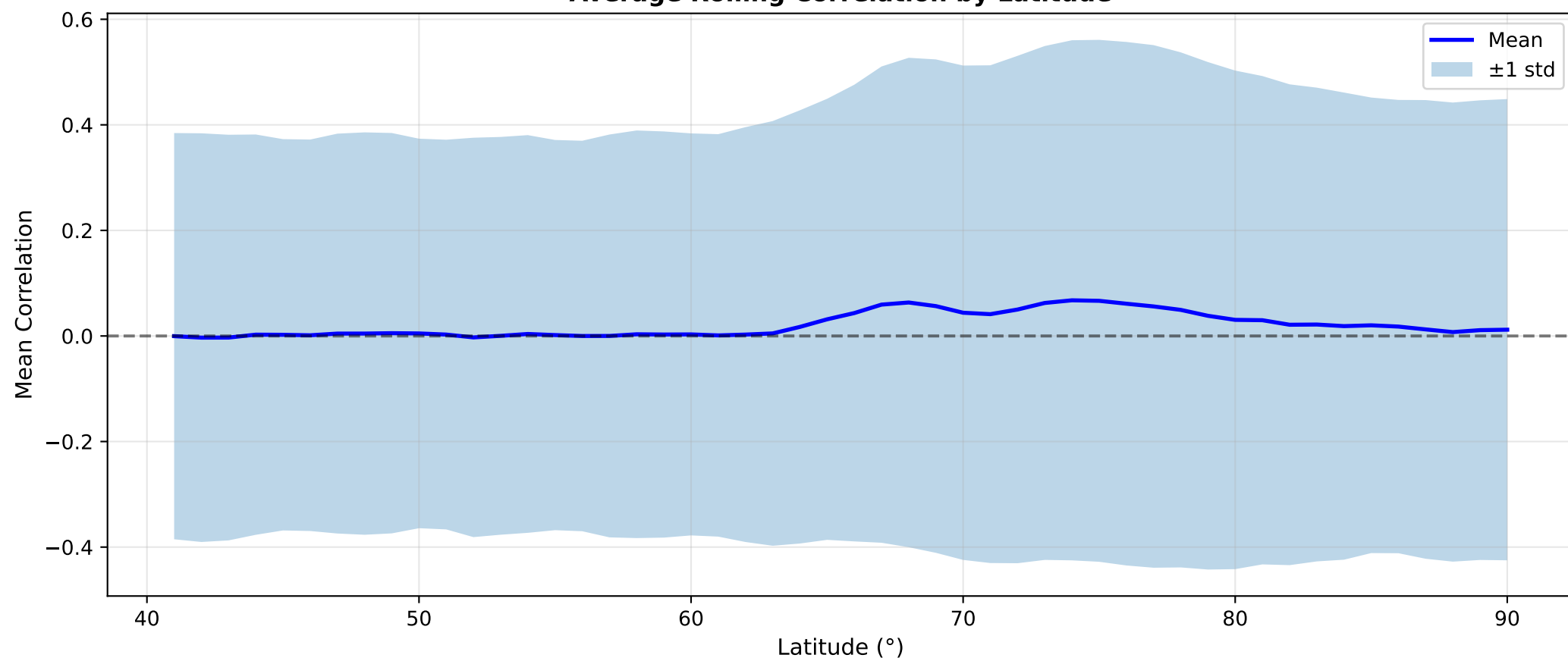
Cross-Correlation Analysis (Lat=70°, MLT 4-8)

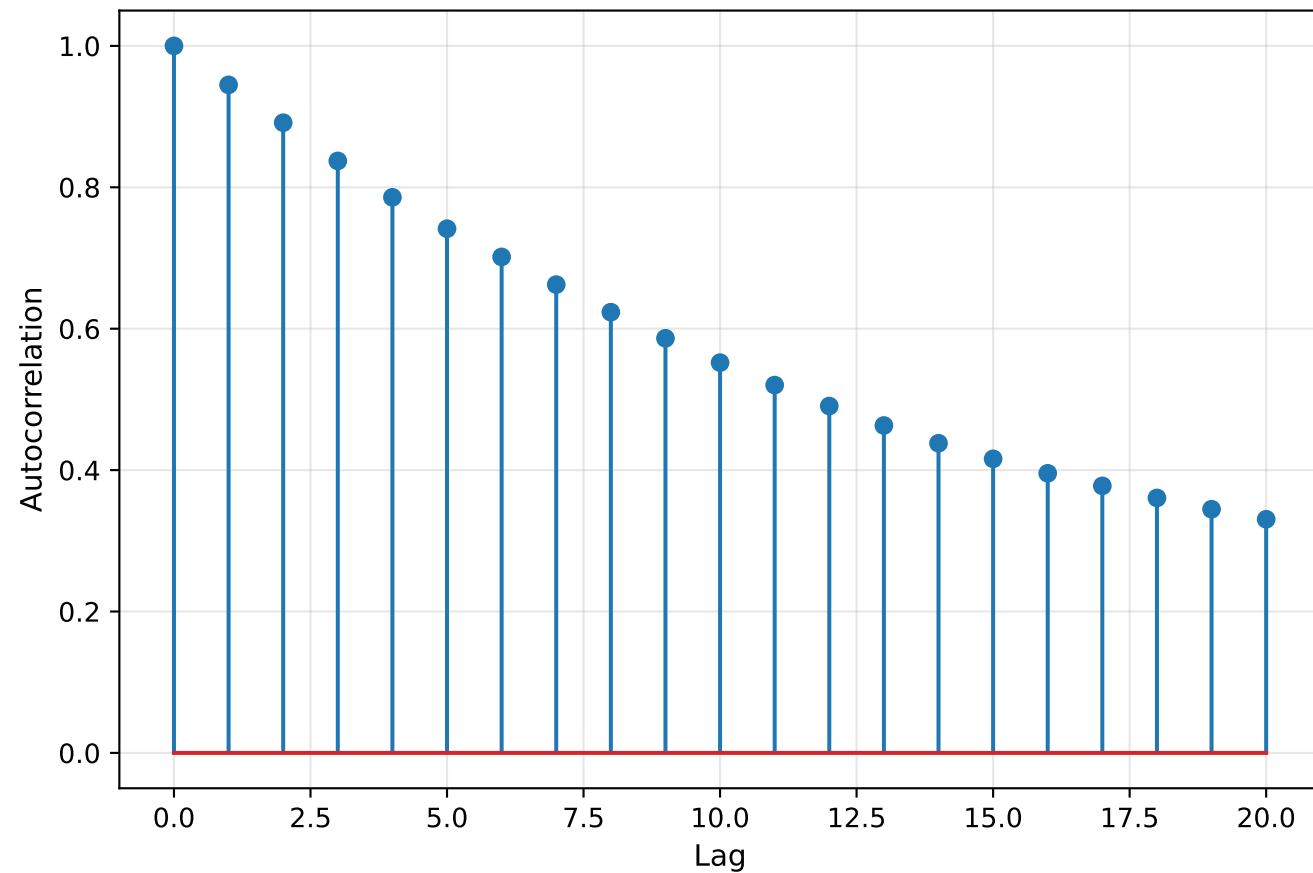
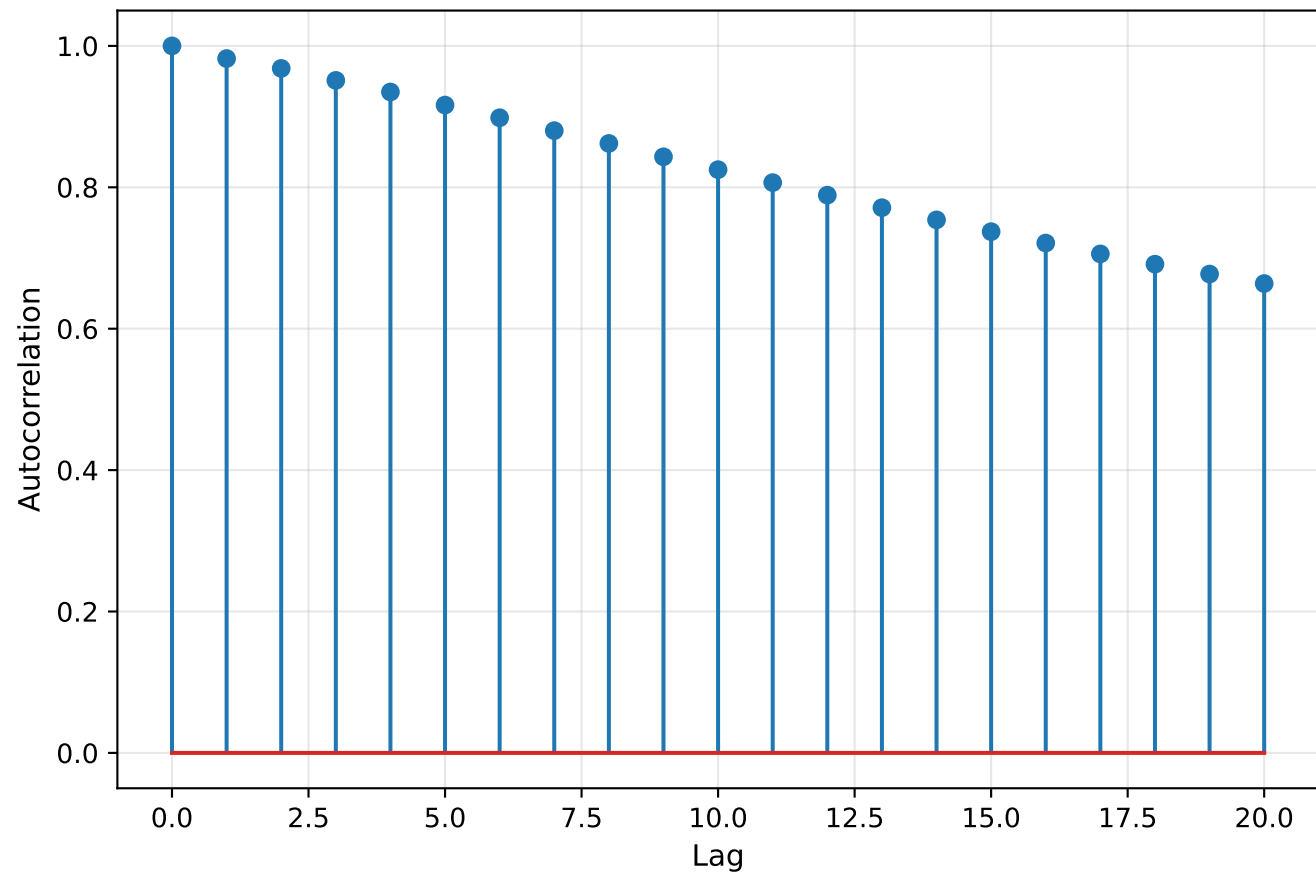


Rolling Correlation (window=10, MLT 4-8)

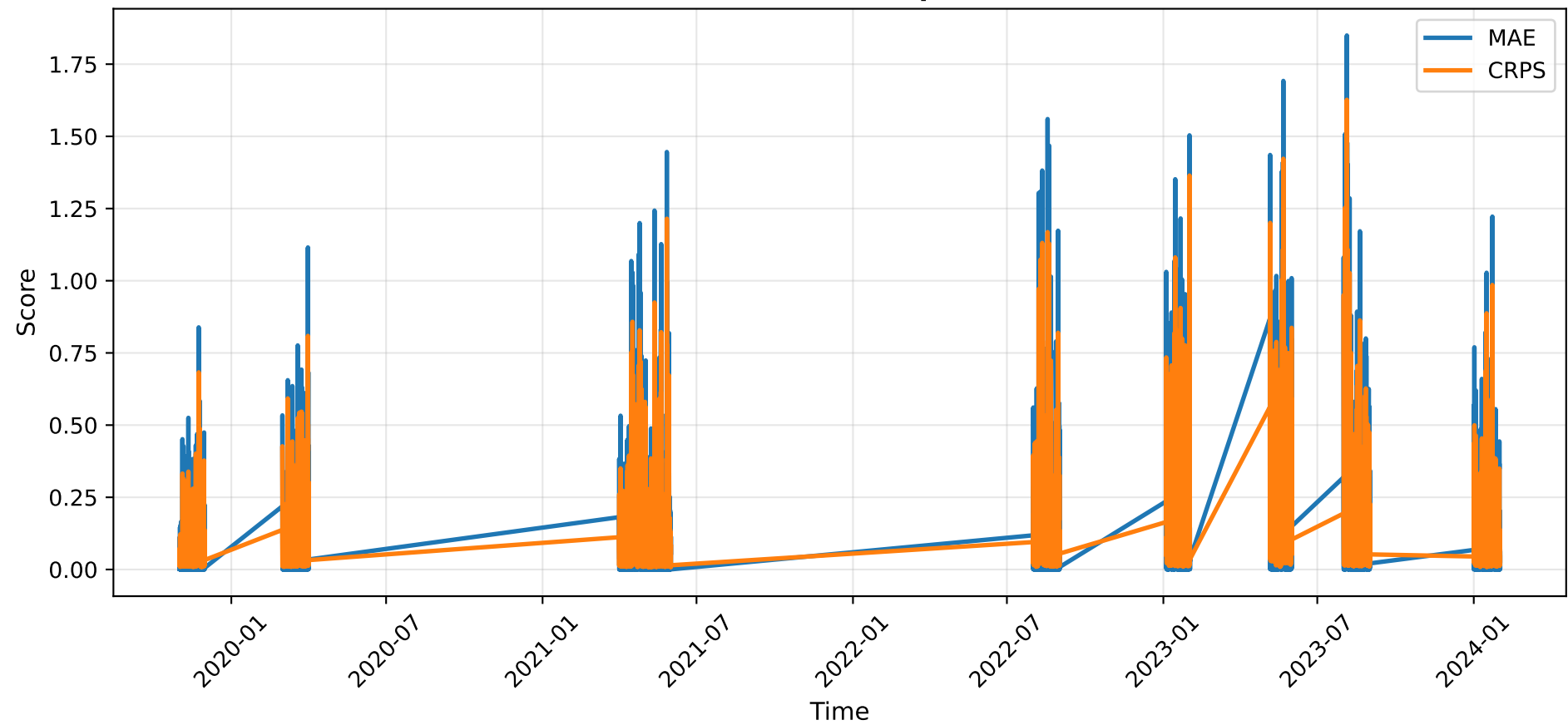


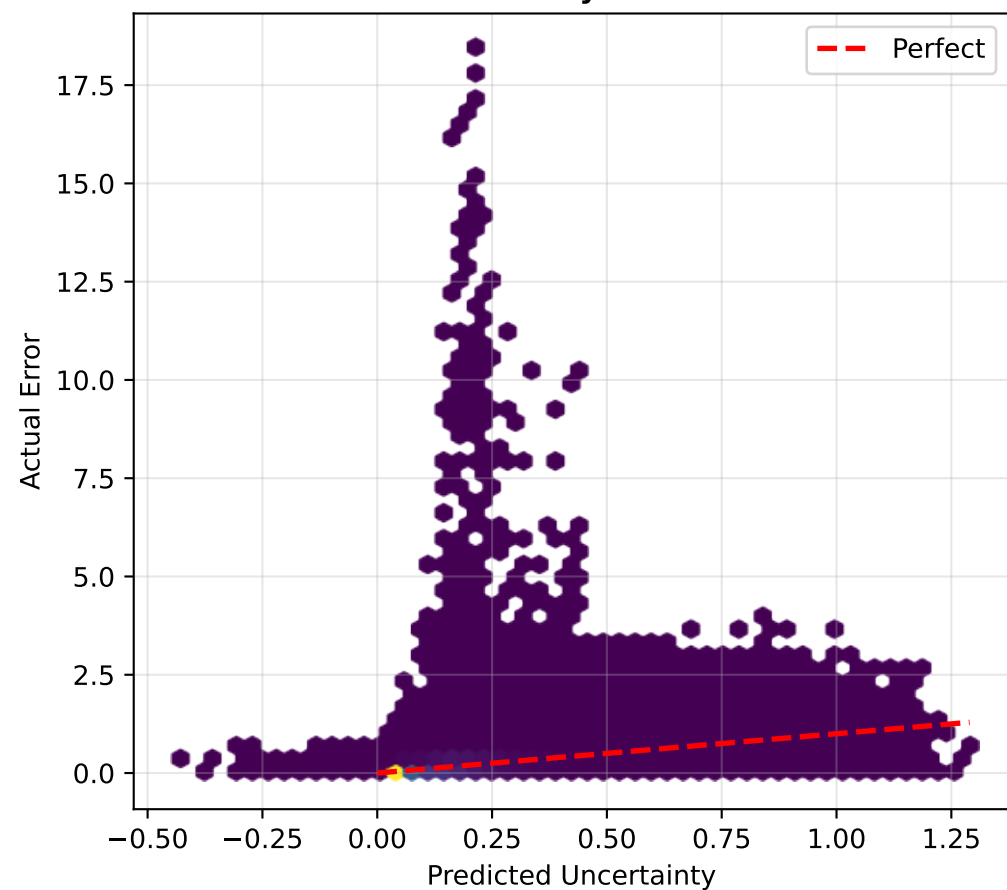
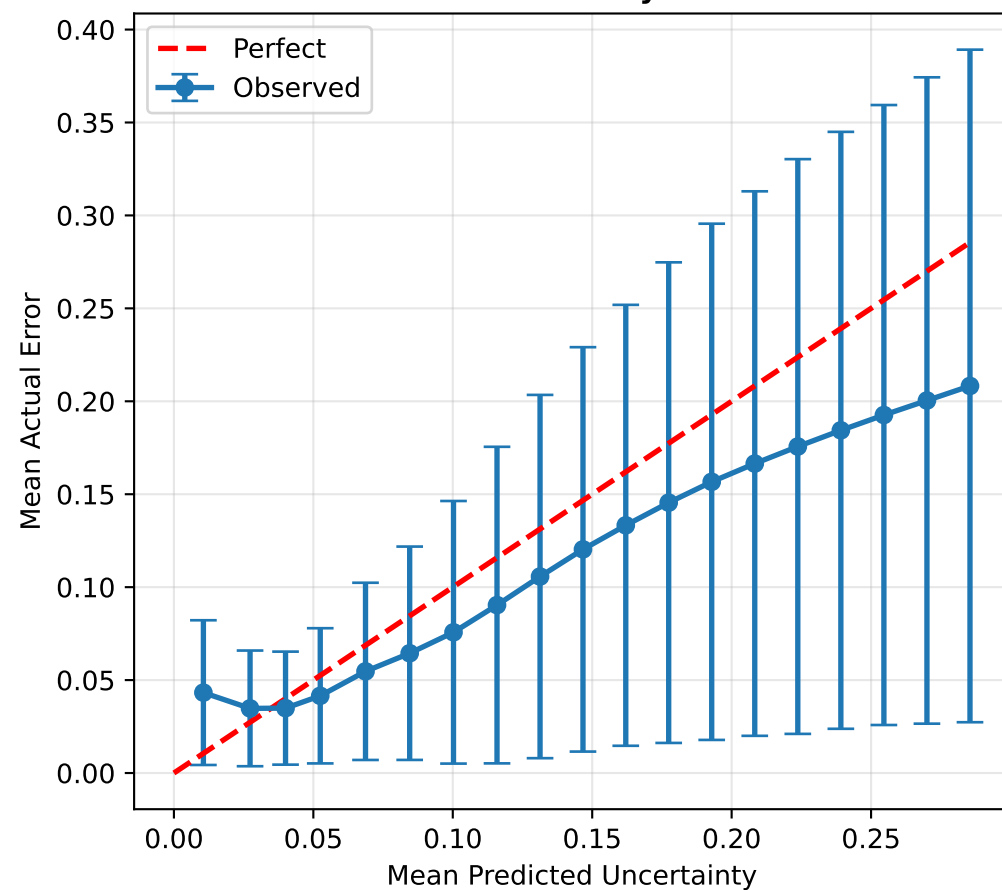
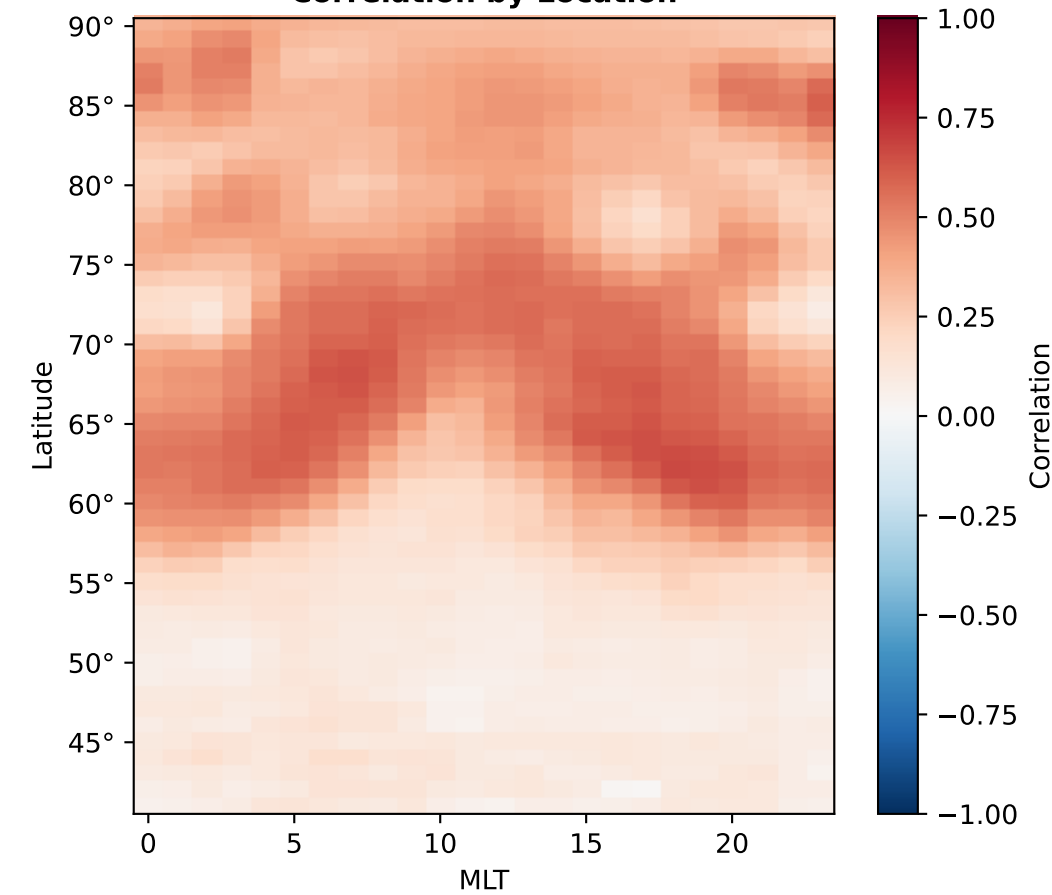
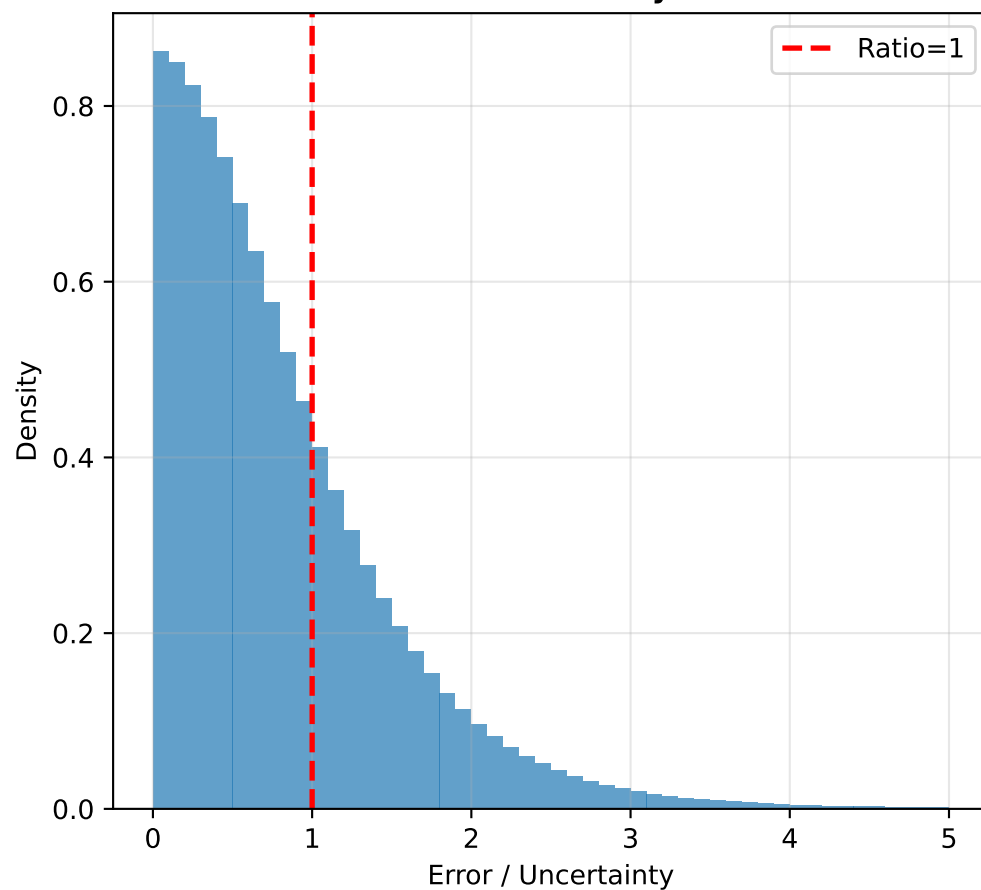
Average Rolling Correlation by Latitude

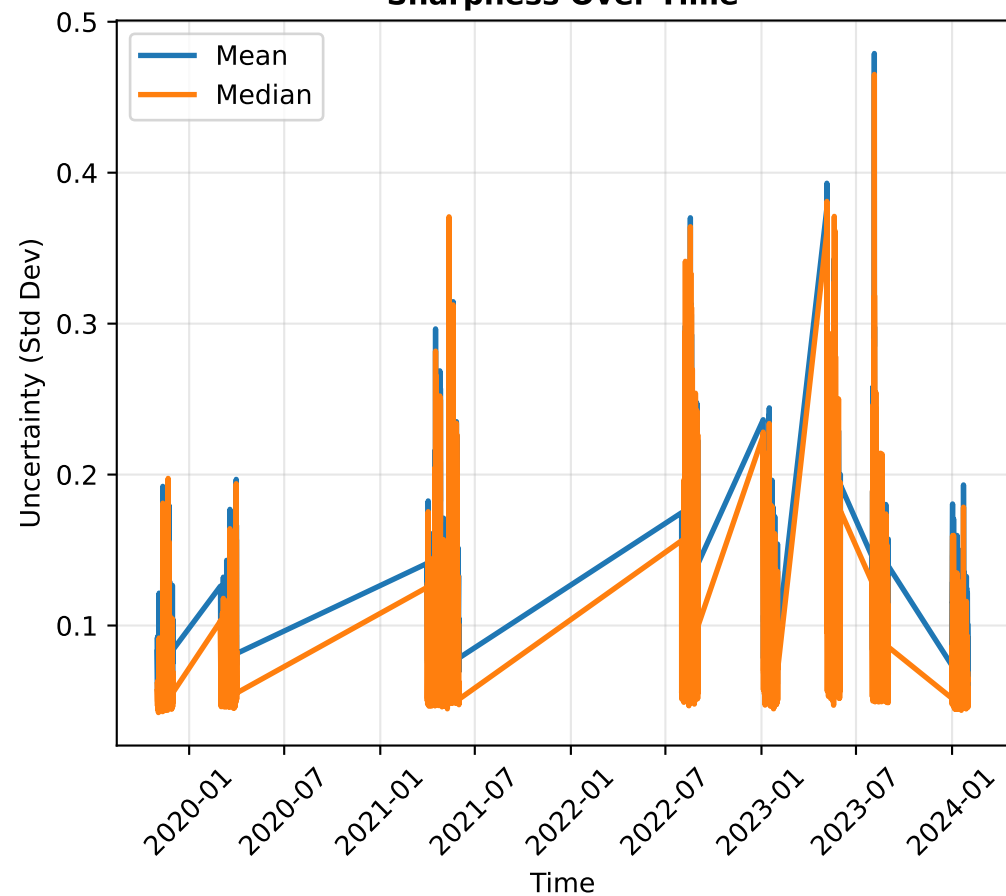
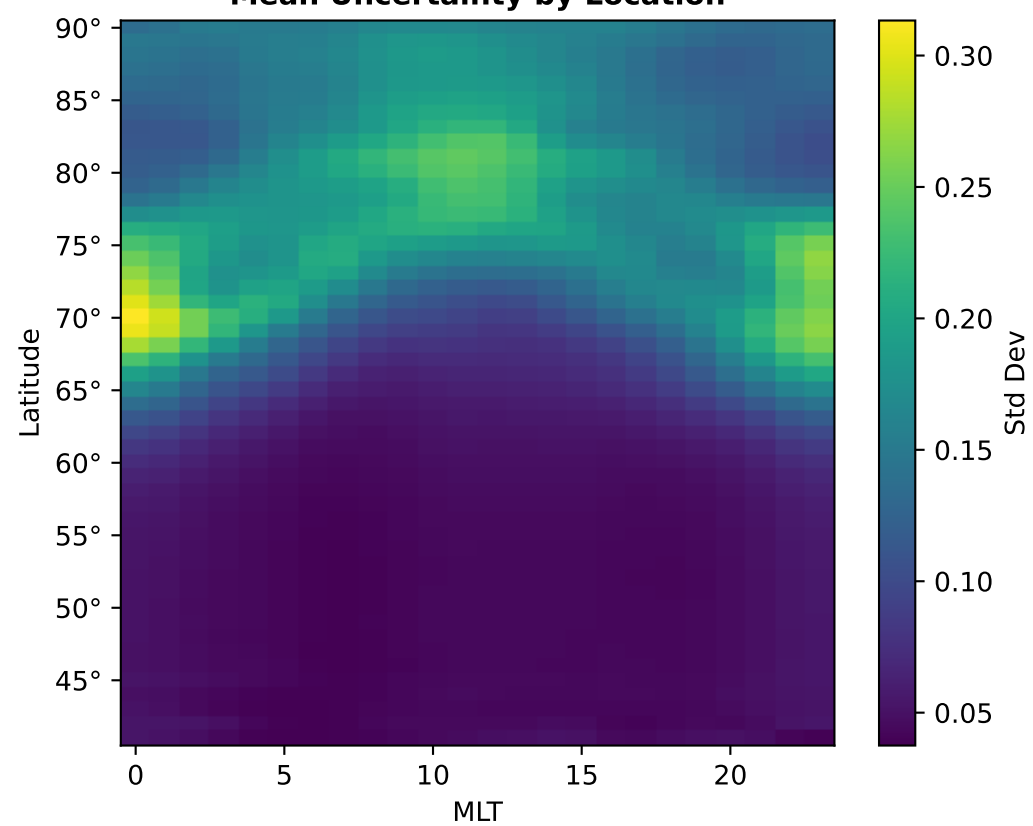
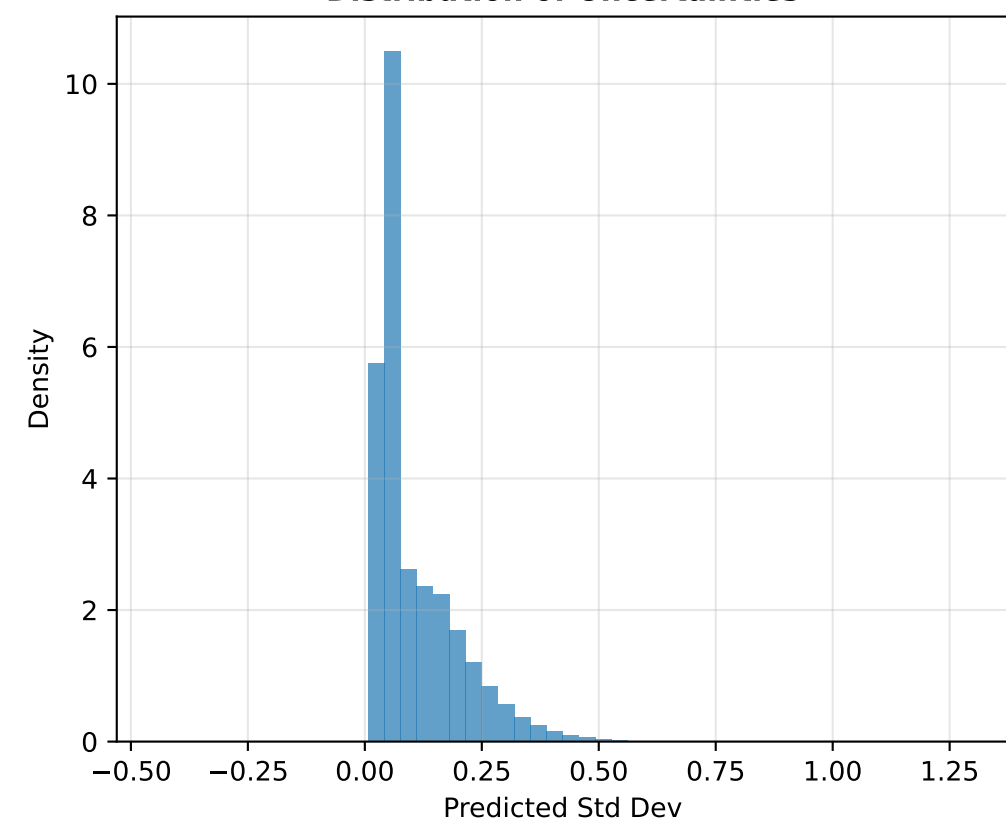
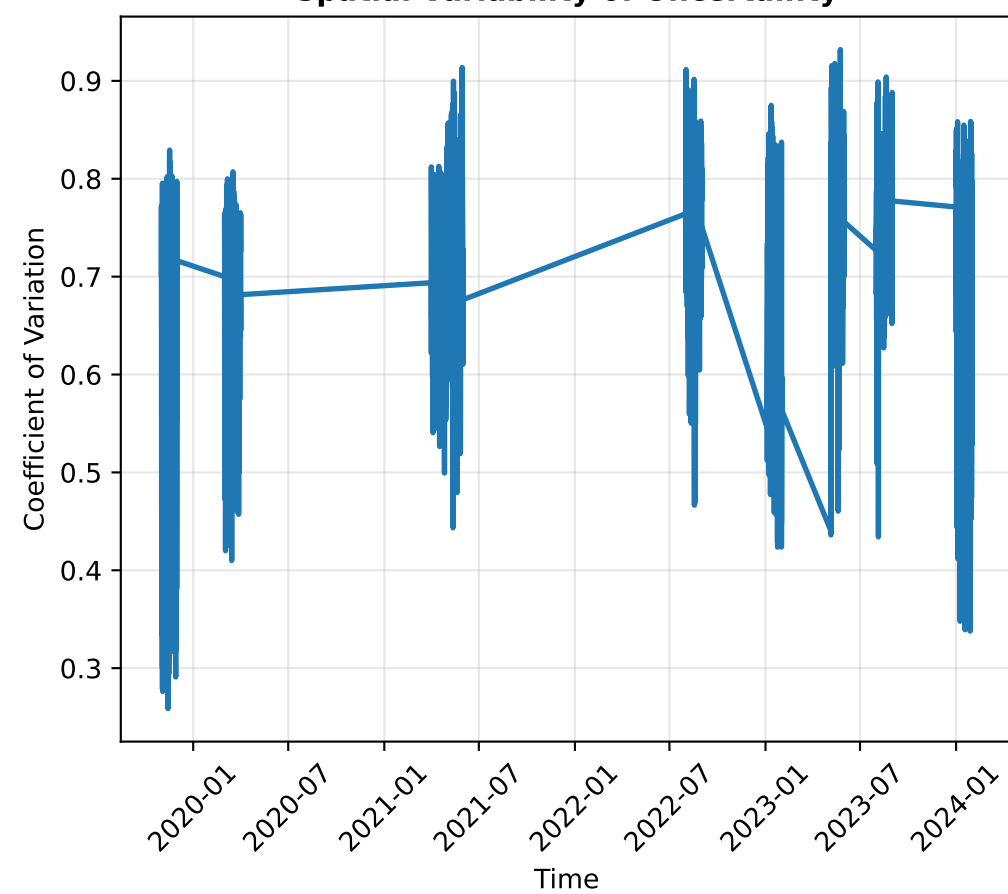


AMPERE Autocorrelation (Lat=70°, MLT 4-8)**Predicted Autocorrelation (Lat=70°, MLT 4-8)**

MAE vs CRPS Comparison



Uncertainty vs Error**Binned Analysis****Correlation by Location****Error-to-Uncertainty Ratio**

Sharpness Over Time**Mean Uncertainty by Location****Distribution of Uncertainties****Spatial Variability of Uncertainty**

Numerical Metrics Summary

Deterministic Performance:

Point Correlation: 0.542

Point RMSE: 0.171

Point MAE: 0.105

Point Bias: 0.032

Uncertainty-Error Corr: 0.591

Executive Summary

Point Performance:

Point prediction at Lat=70°, MLT 4-8 shows moderate correlation ($r=0.542$) with RMSE=0.171. Model exhibits positive bias of 0.032. Uncertainty-error correlation is 0.591, indicating strong awareness of prediction quality.

Spatial Performance:

Spatial analysis at MLT 4-8 shows mean RMSE of 0.092 across all latitudes. Maximum errors (0.211) occur at high latitudes.

Temporal Alignment:

Model leads observations by 1 timesteps (anticipates changes). Maximum achievable correlation is 0.544.

Time-Varying Performance:

Rolling correlation analysis (window=10) shows mean correlation of 0.021 with range [-0.997, 0.999]. Stable performance across time.

Temporal Dynamics:

Autocorrelation analysis shows decorrelation timescales of 18 timesteps for observations and 20 timesteps for predictions. Model captures temporal persistence well.

Uncertainty Quality:

CRPS/MAE = 0.73; Uncertainty reduces error by 26.6% compared to just using the mean. Mean CRPS=0.0773, Mean MAE=0.1053.

Uncertainty-error correlation is 0.567, indicating weak relationship. Median error/uncertainty ratio is 0.639 (underconfident (uncertainties too large)). Model shows moderate awareness of uncertainty patterns.

Uncertainty Characteristics:

Model sharpness: mean uncertainty=0.1073, median=0.0623. Temporal variability (CV=0.365) suggests condition-dependent uncertainty. Spatial variability (CV=0.611) indicates location-dependent confidence.